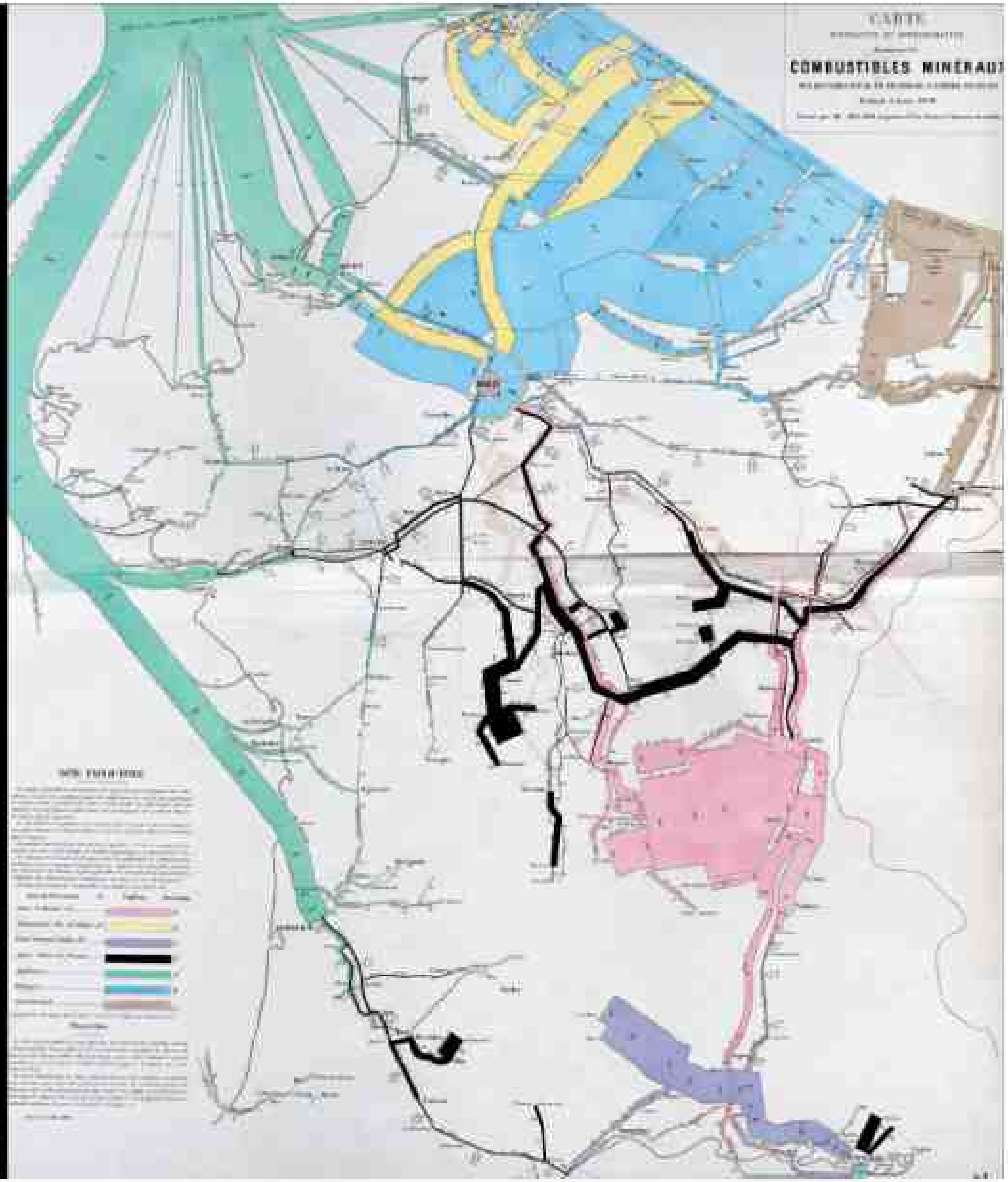
Geographical Data and Geographic Information

GEOG 1026
Department of Geography
National Taiwan University



Figurative map of mineral fuels movement on the waterways and iron of the French Empire in the year 1856, 1858, Charles Minard

Last Week

- What is geography?
  - Four traditions of geography
  - Physical and human geography
  - Quantitative and qualitative geography
  - Geographic concept
  - Laws of geography

Course Objectives

- Maps are the language of geography. Cartography is the science, technique and art of creating maps. It involves collecting geographic information, selecting and simplifying data, and producing maps that communicate spatial patterns, relationships, and geographic phenomena.

- Geographic information systems (GIS) are computer-based systems to collect, model, process, analyze, and visualize geographic information.

- This course integrates theory and practice by introducing the principles of cartography and the theoretical foundations of GIS, strengthening spatial thinking, and cultivating students' ability to explore geographic issues using geographic information.

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This Week

- Spatial data, geospatial data, and geographic data
- Geographic information
- Geographic thinking

- Spatial data 空間資料

- Geospatial data 地理空間資料

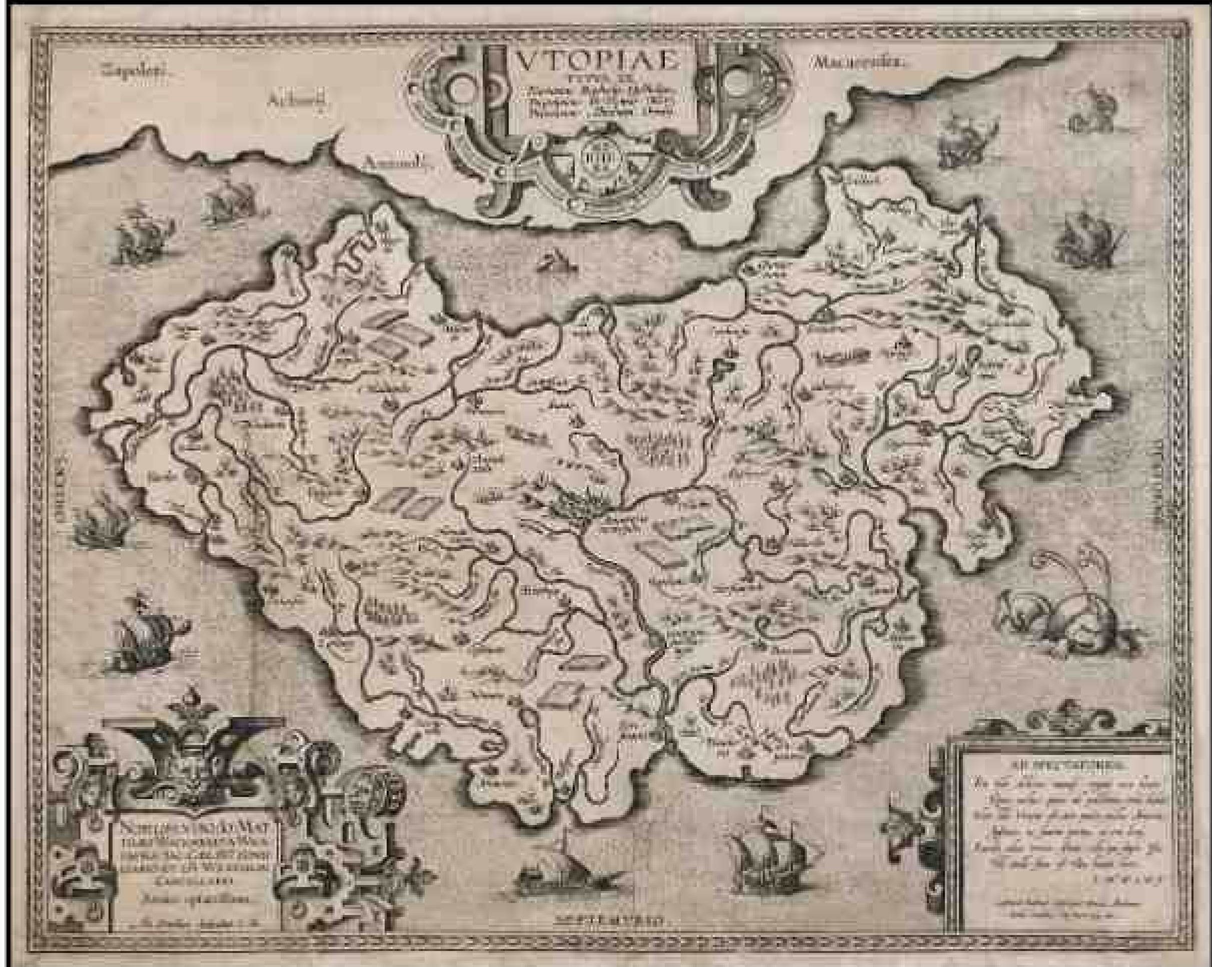
- Geographic data 地理資料

What is spatial data?

Spatial Data

- Data that has a “spatial” component
- Data is connected to a coordinate system
- Represent the location, size, and shape of objects
- May include more attributes providing other information

Character Profiles
Adventure Log
L
Map
R
Album
Hyrule Compendium
Akkala
Hebra
Central Hyrule
Stamp/Pin
Lanayru
Necluda
44.58/100%
3:35 PM
77 °F
1063, 1281, -0495
Shrine of Light
Switch
Move
Zoom
Hero's Path Mode
Sensor
Select
Return



MAP of Utopia, 1595-1596, Abraham Ortelius

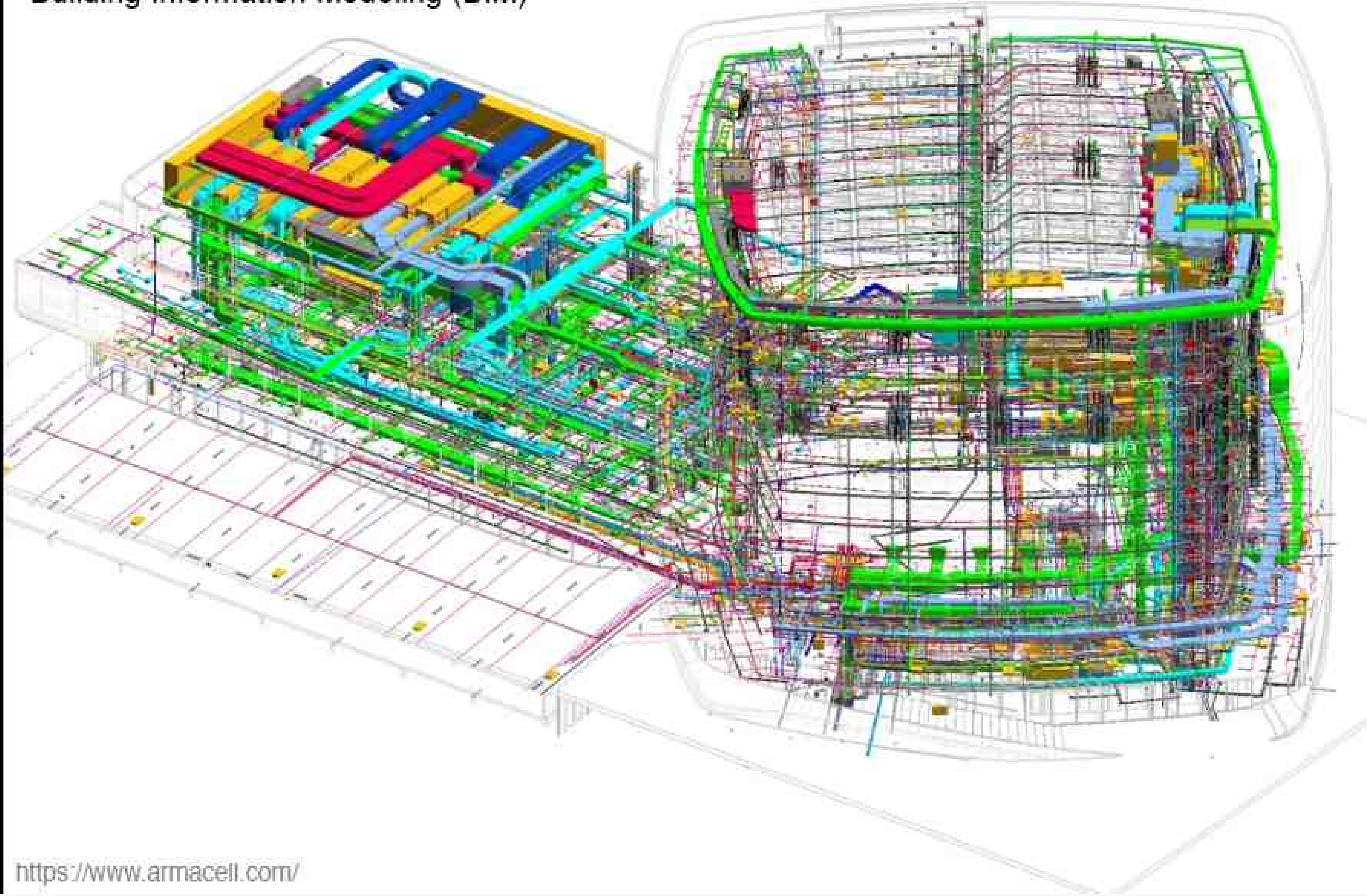
Lunar Landings Map, 1969, NASA

What is geospatial data?

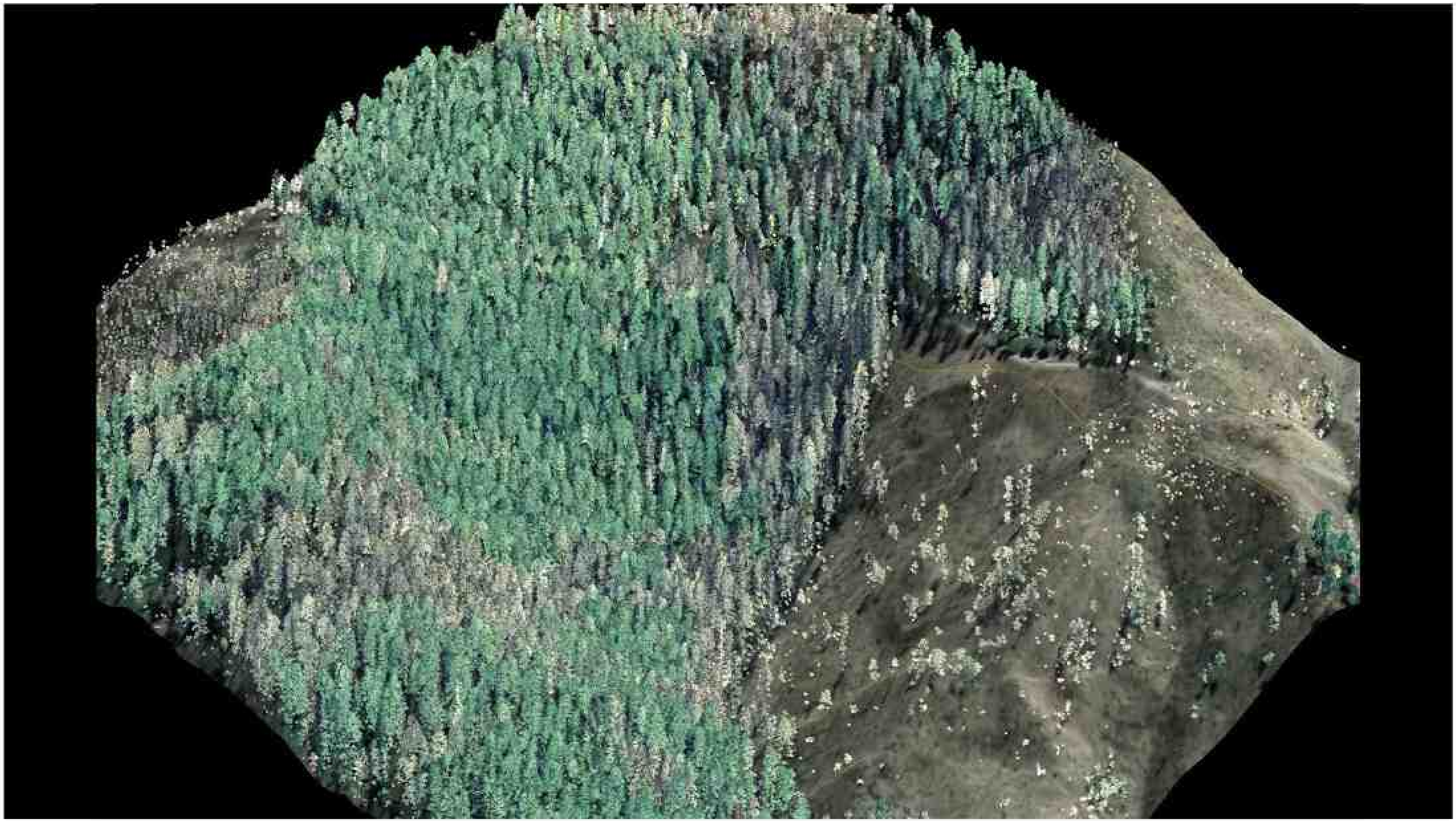
Geospatial Data

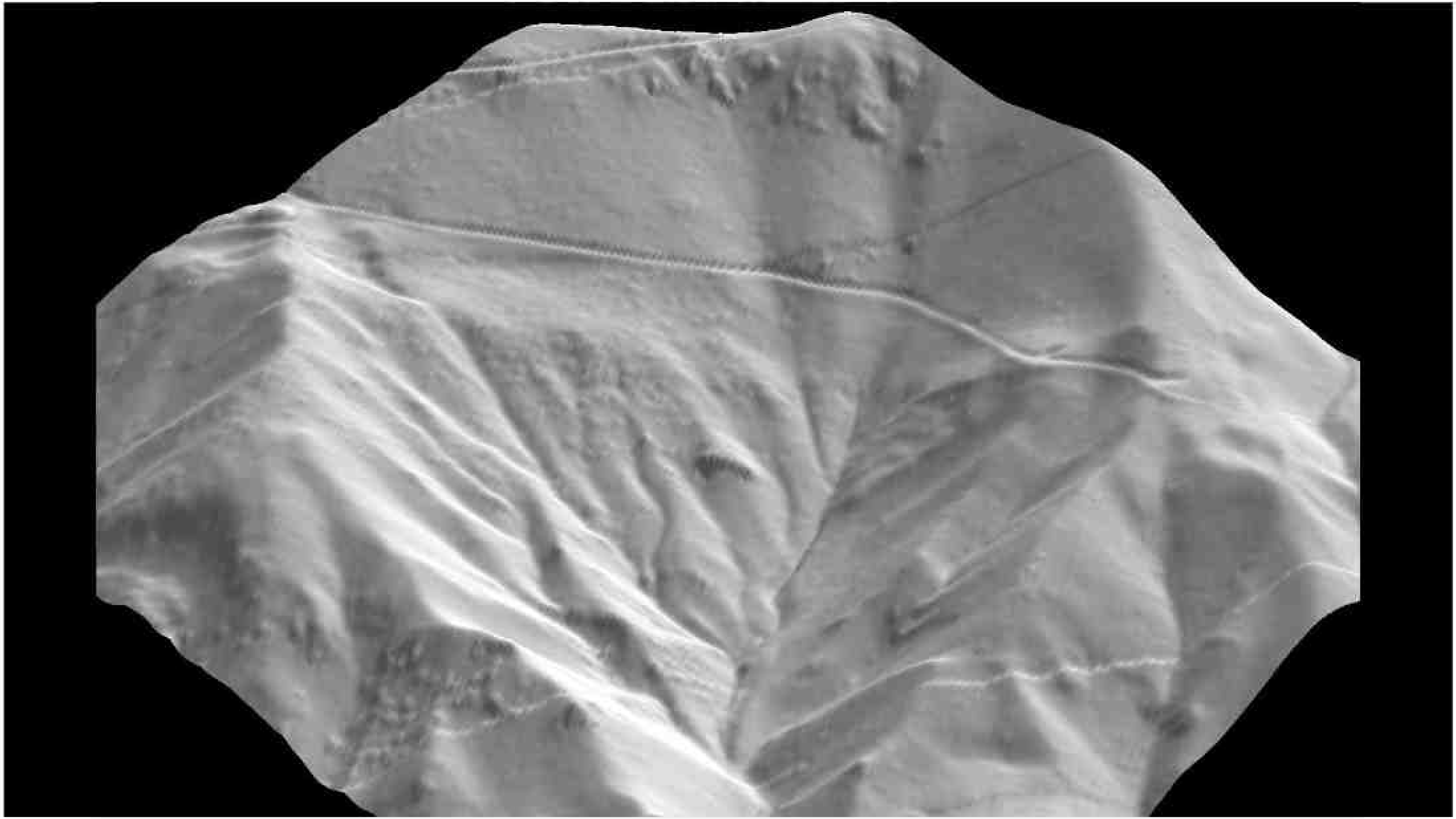
- Data that has a “spatial” component
- Data is connected to a place in the Earth
- Numerical/digital values in GIS represent physical objects
- Represent the location, size, and shape of physical objects
- May include more attributes providing other information

Building Information Modeling (BIM)



https://www.armacell.com/

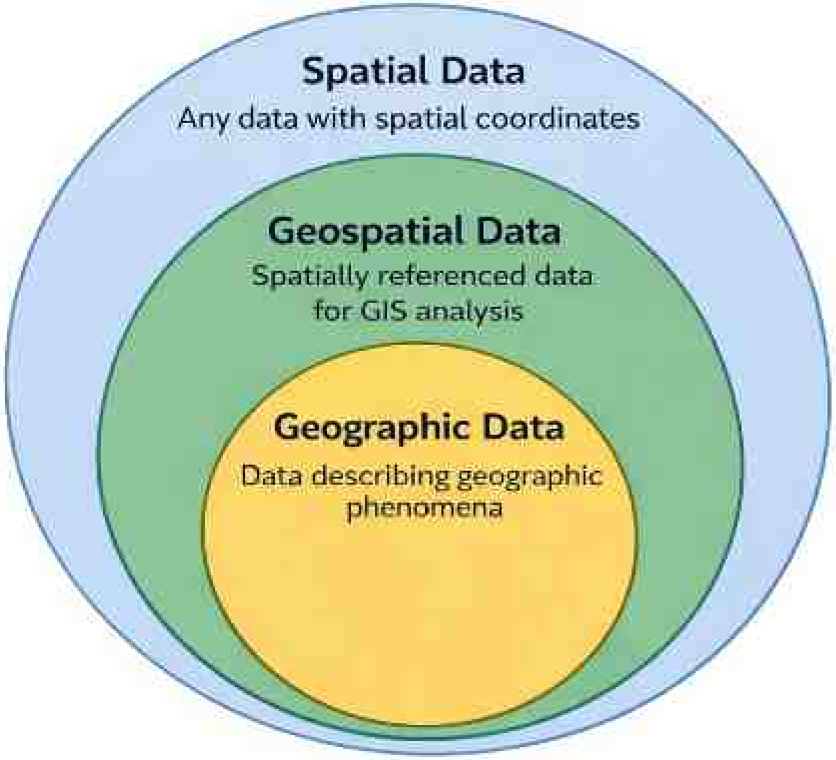




What is geographic data?

Geographic Data

- Data that has a “spatial” component
- Data is connected to a place in the Earth
- Numerical/digital values in GIS represent physical objects
- Represent the location, size, and shape of physical objects
- May include more attributes providing other information
- Describe geographic phenomena related to physical and human geography



- **Spatial data:** Data with spatial coordinates (e.g., x, y, z) in any context
- **Geospatial data:** Spatial data specifically linked to locations on Earth's surface; used for GIS analysis and geospatial technologies
- **Geographic data:** Geospatial data that describes geographic phenomena, features, or conditions

Why is the map/bird-eye view good for geospatial data?

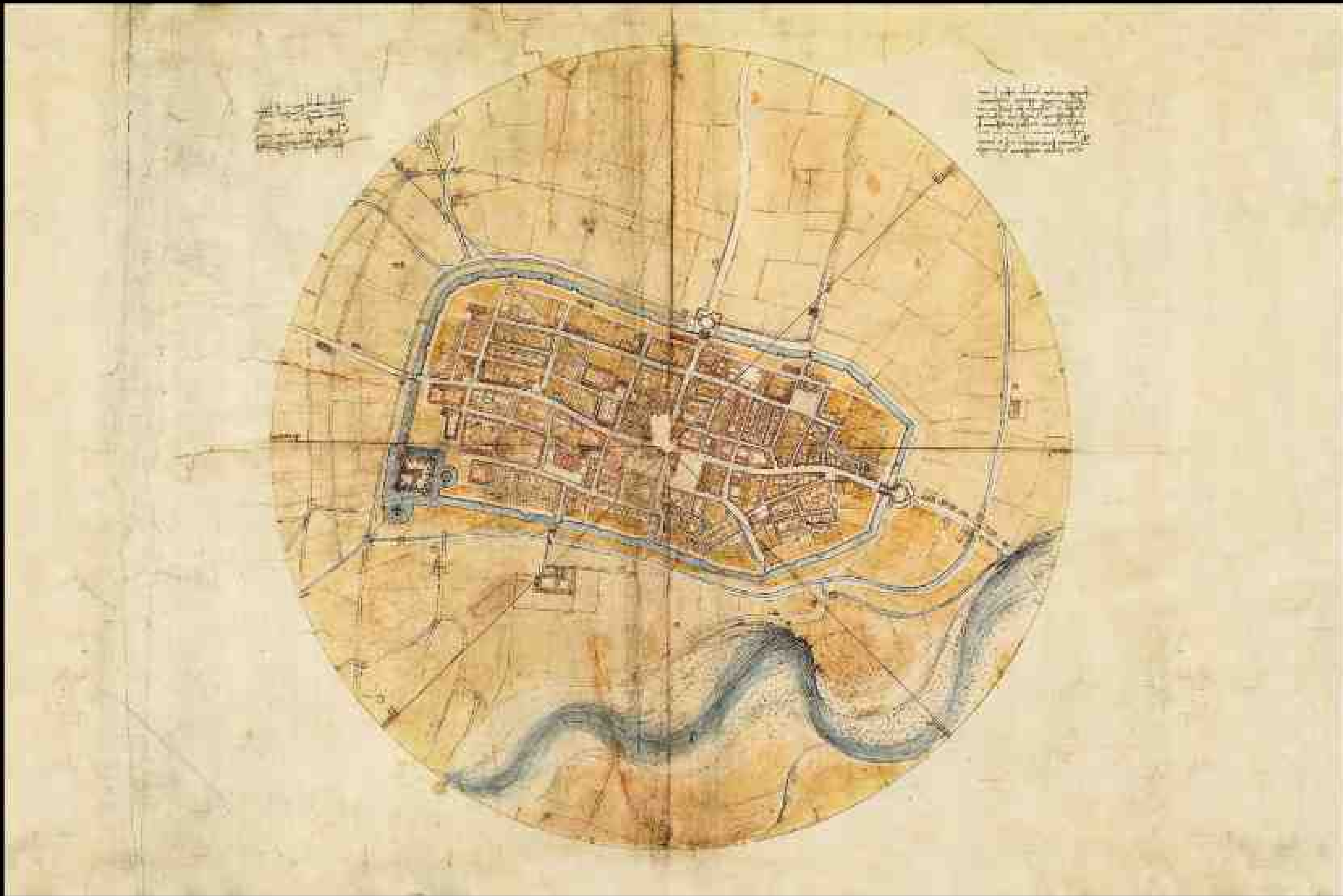

看見・齊柏林基金會
Chi Po-lin Foundation
花蓮玉里 大腳印 | 齊柏林空中攝影


看見・齊柏林基金會
Chi Po-lin Foundation
玉山頂上 原聲童聲合唱團 | 齊柏林 空中攝影

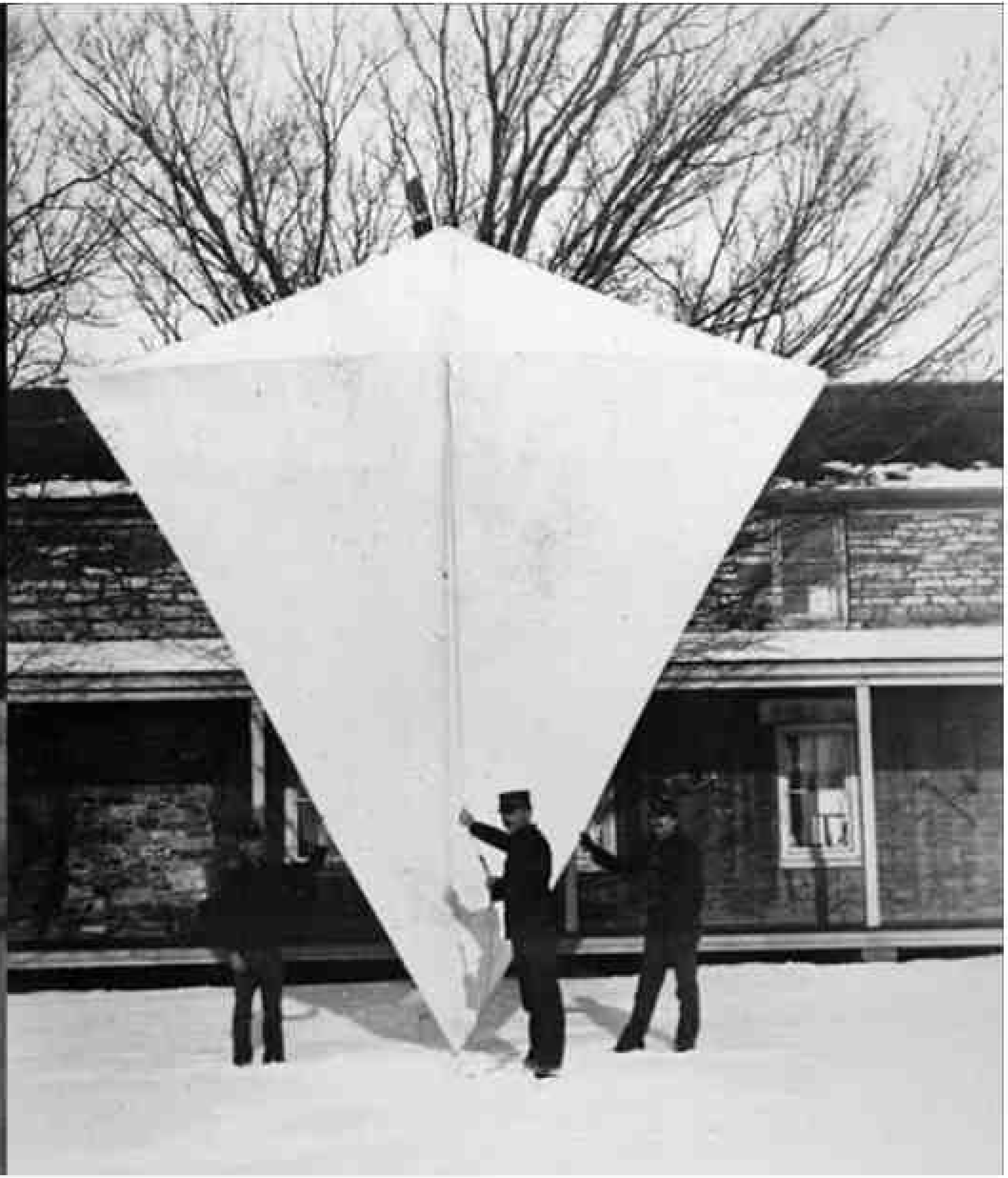

看見・齊柏林基金會
Chi Po-lin Foundation


看見・齊柏林基金會
Chi Po-lin Foundation
屏東恆春 梅花鹿 | 齊柏林空中攝影

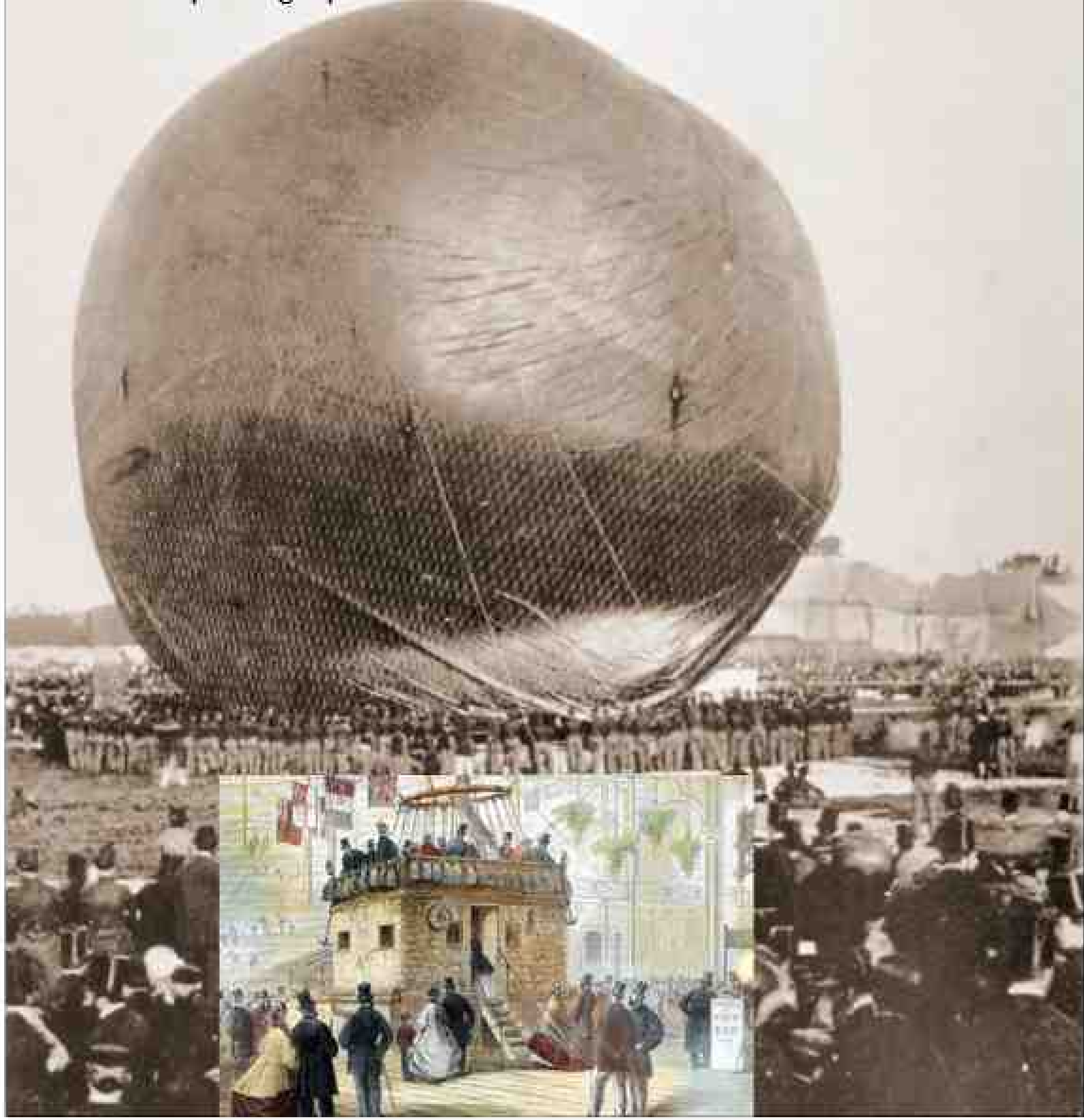
Taioan, Formosa and Fort Zeelandia from Atlas Blaeu-Van der Hem – Taioan, 17th century

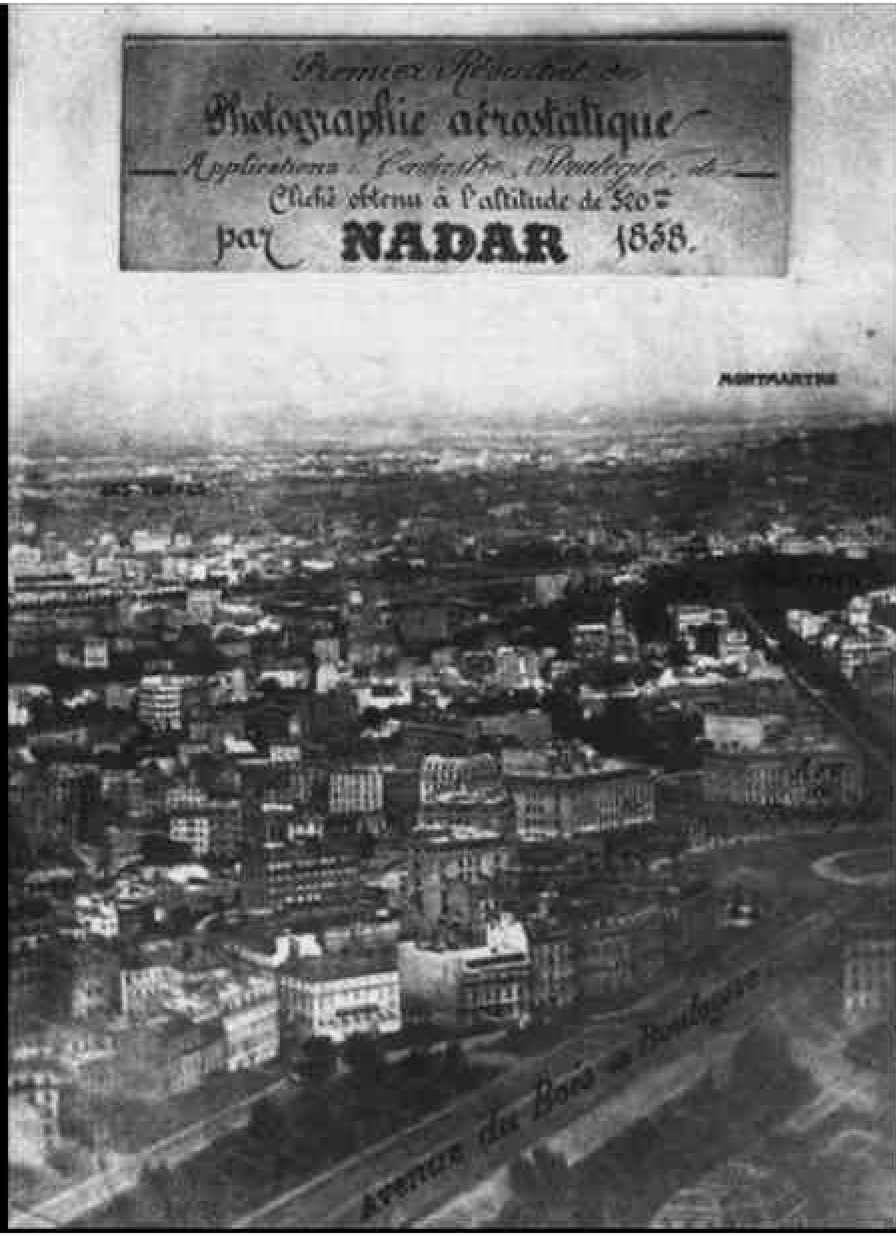


Map of Imola, 1502, Leonardo da Vinci



First aerial photographer: Nadar in 1858





Compagnie d'aérostiers at Battle of Fleurus

Civil war ballooning

Aerial photography in WWI

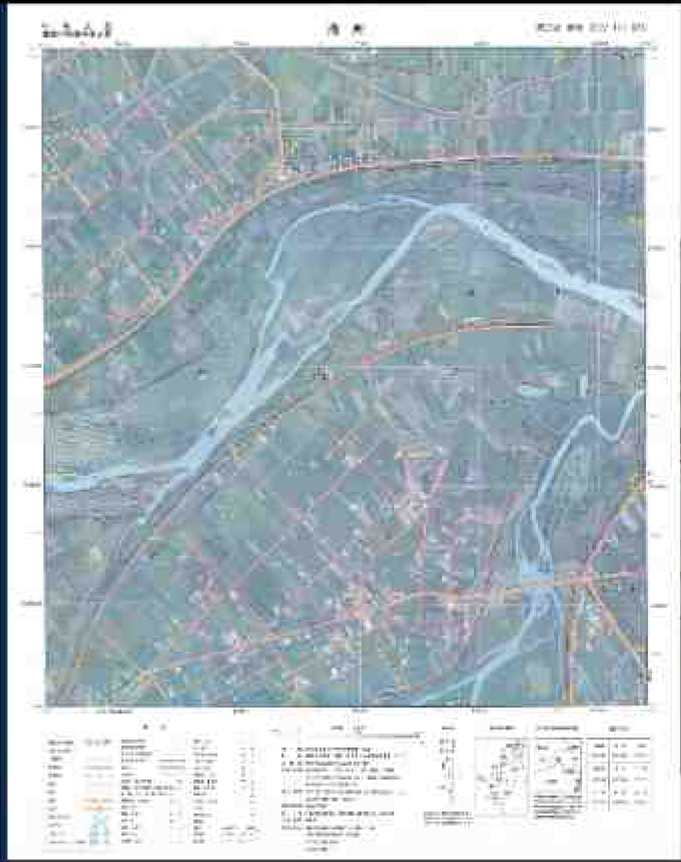

6
https://www.gjtaiwan.com/oldtwcolor


War Theatre #11 (Formosa) - BOMBING (over)
OKAYAMA
Print rec'd Dec 44 from Pub., Sec., 10/AS Intell., Used in Dec 44 issue of Impact. Copied 12/14/44. Used in Press Release ATTACK ON OKAYAMA AIRCRAFT PLANT, FORMOSA, 12/27/44.
Aircraft plant

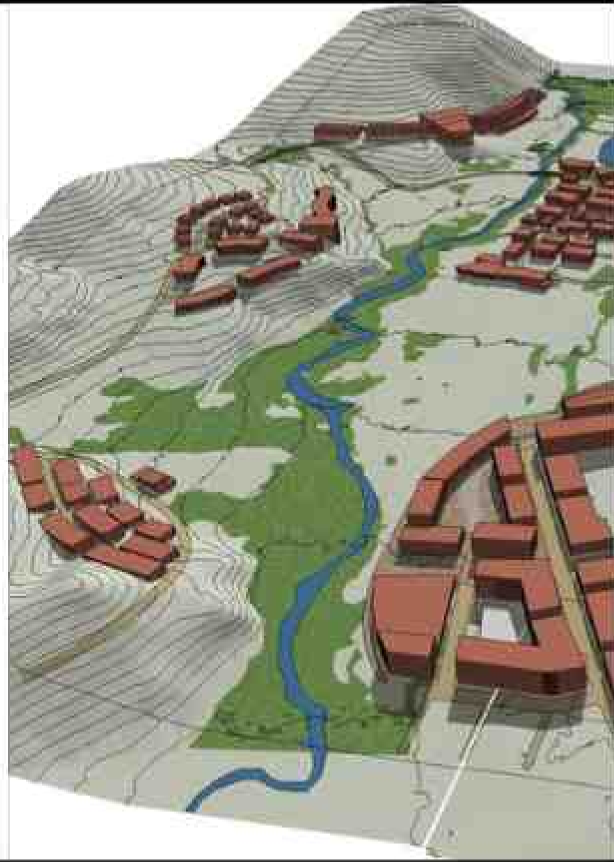
Geospatial Data Presentation

Remote sensing imagery

Topographic map

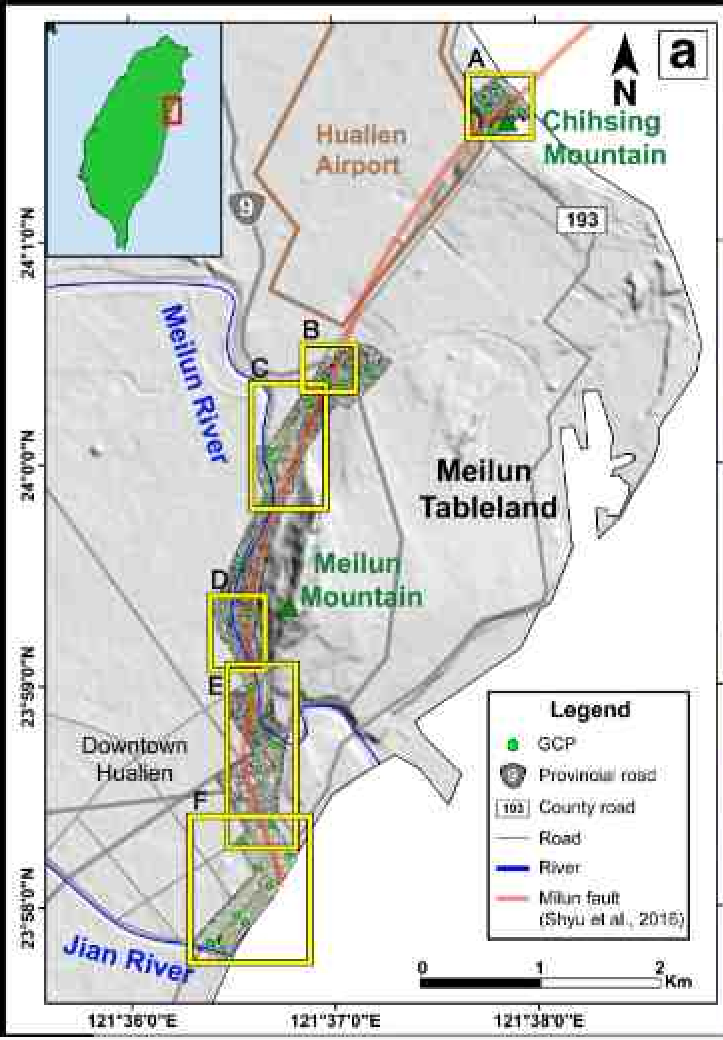


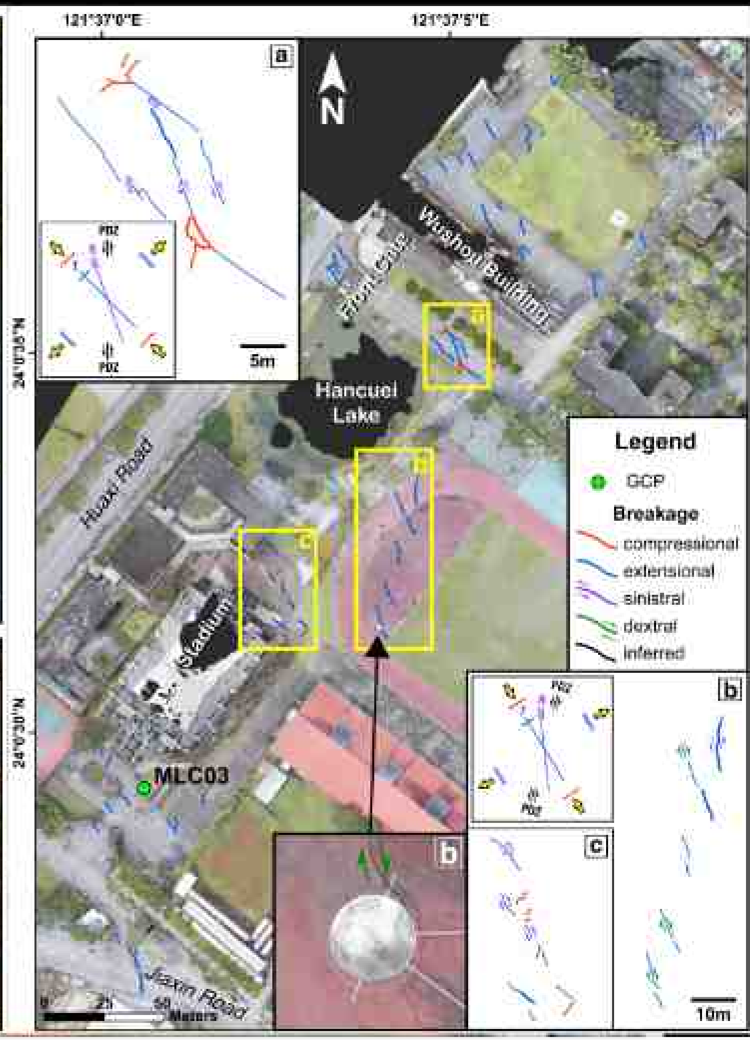
3D model



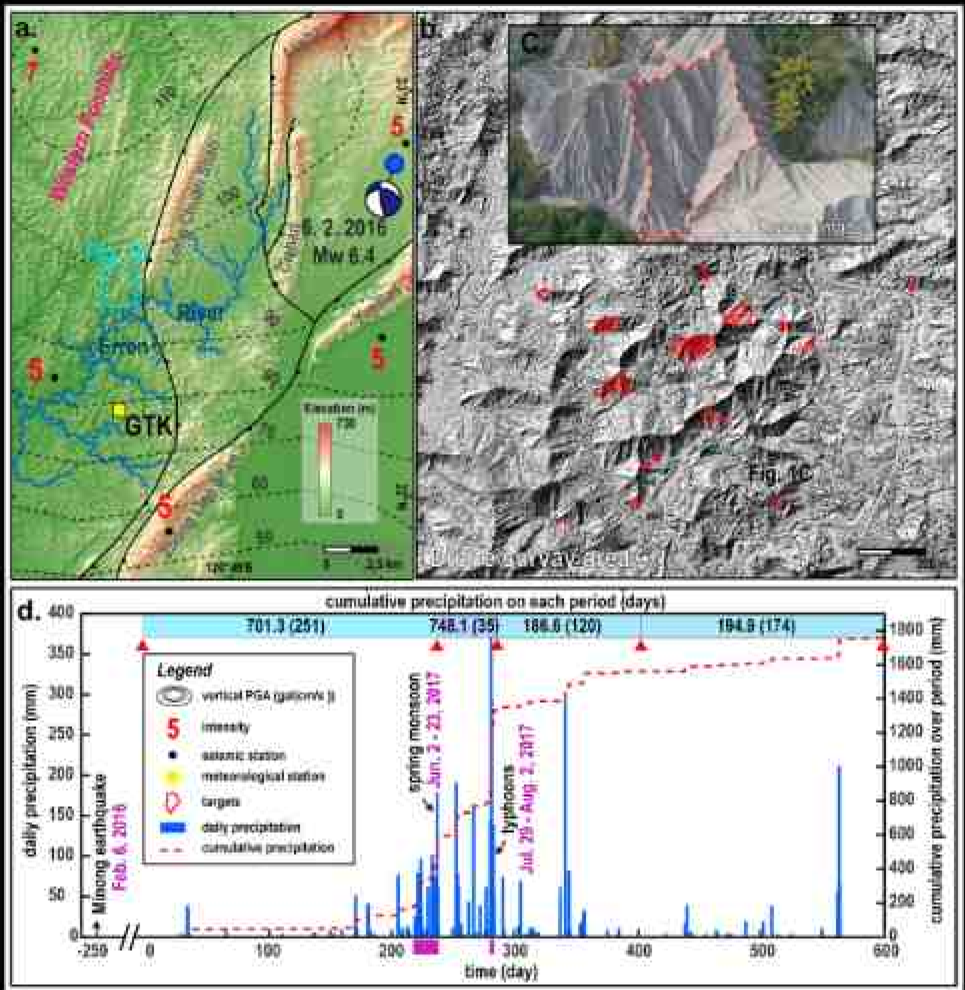
Why do we need geospatial data?

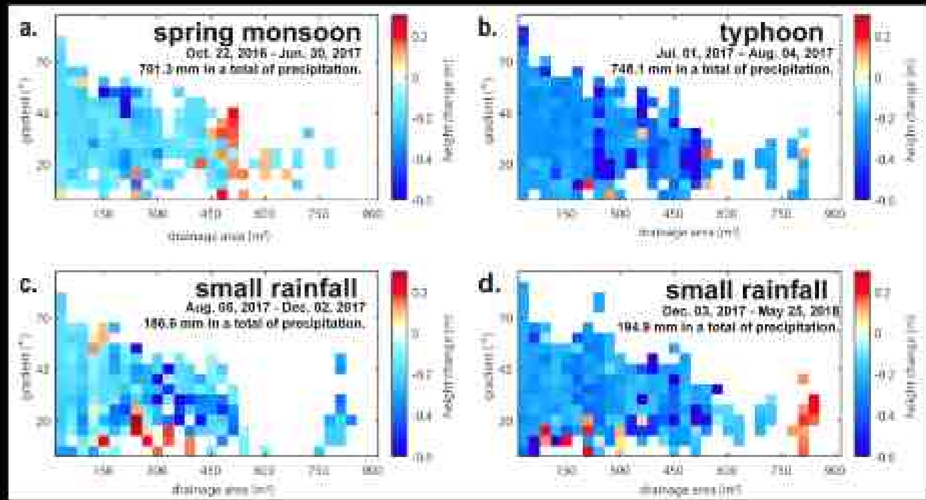
Tectonic Geomorphology and Earthquake Hazards

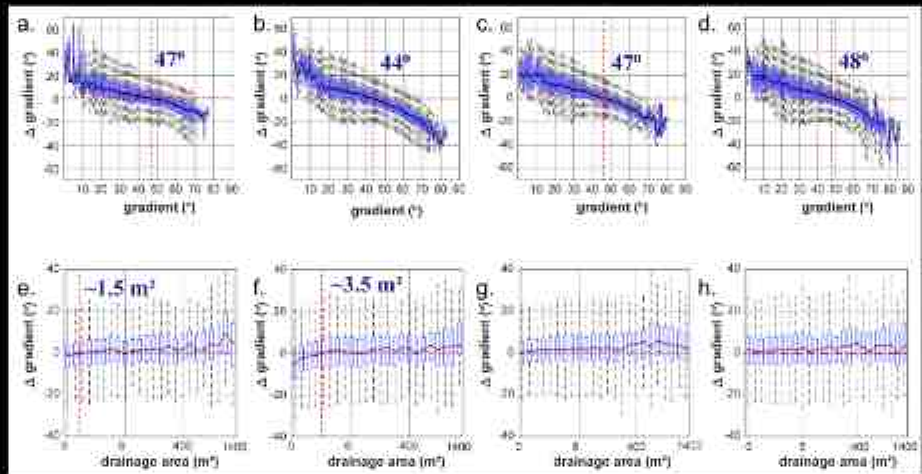




Hillslope-Fluvial Processes

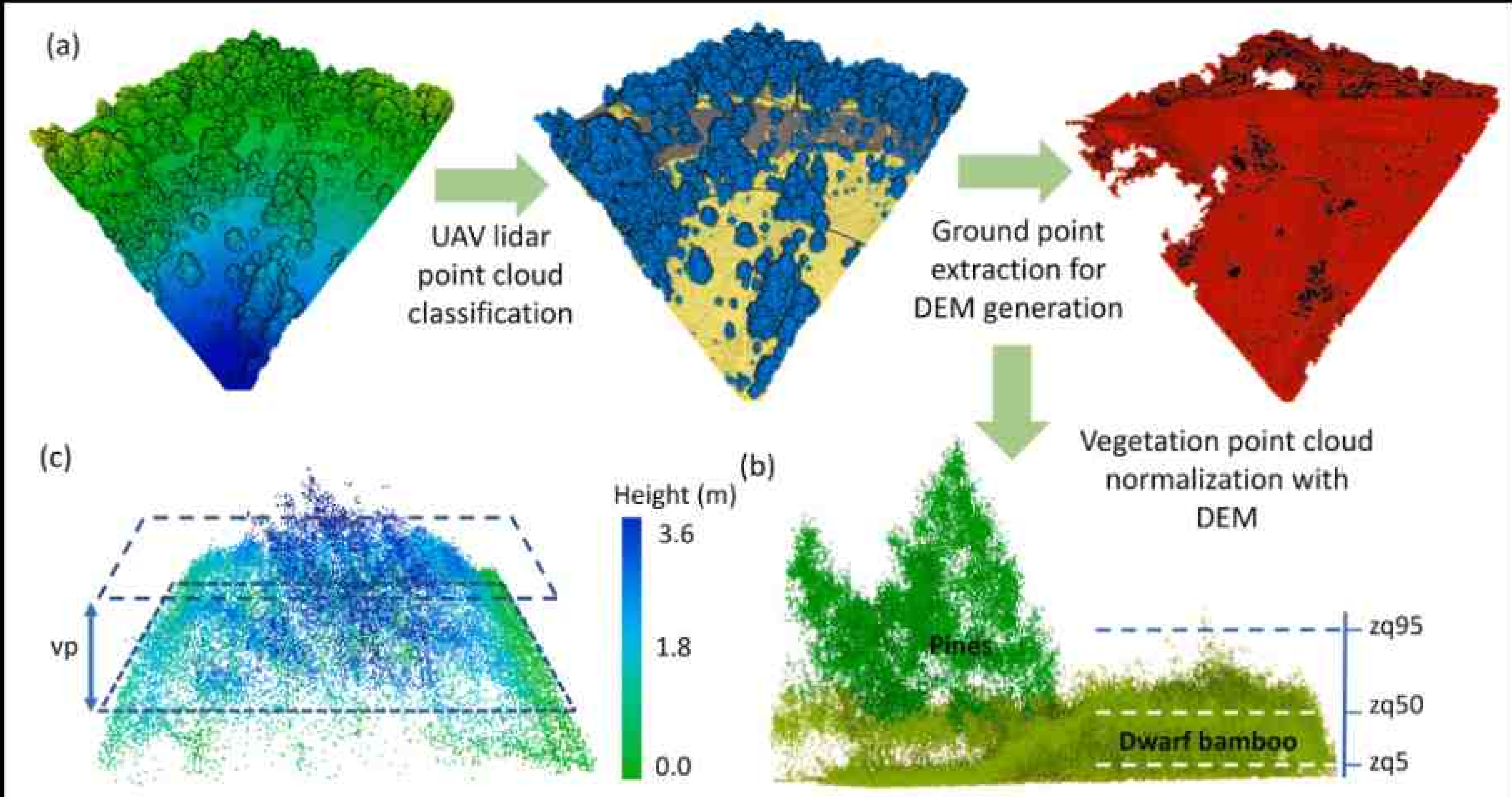






Yang et al. (2021)

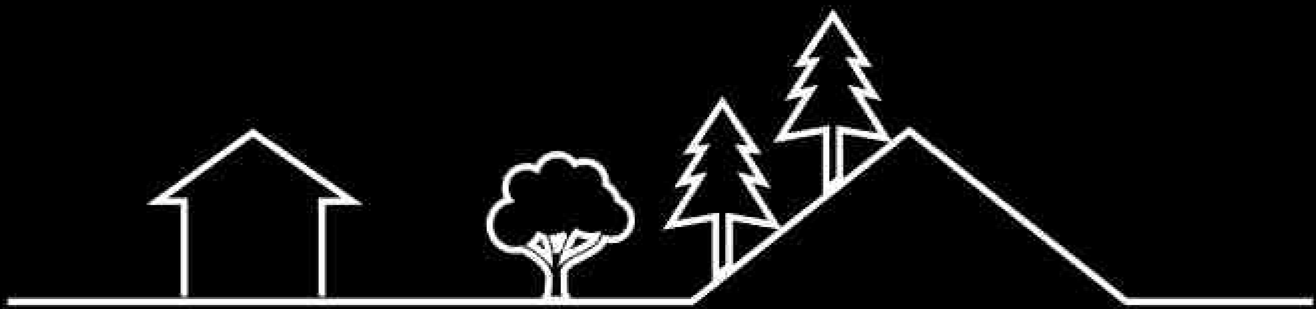
Carbon Cycle and Carbon Storage



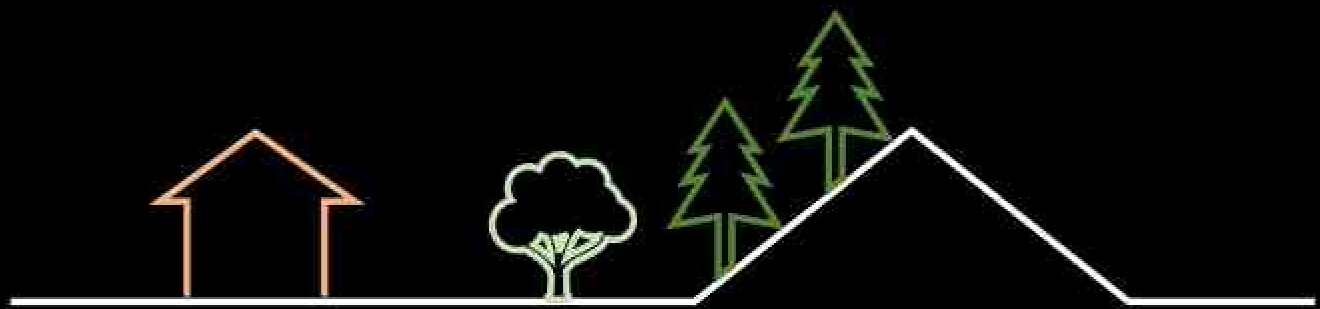
Pan et al. (2023)

Urban Development and Inequality

Spatial Data for Geography



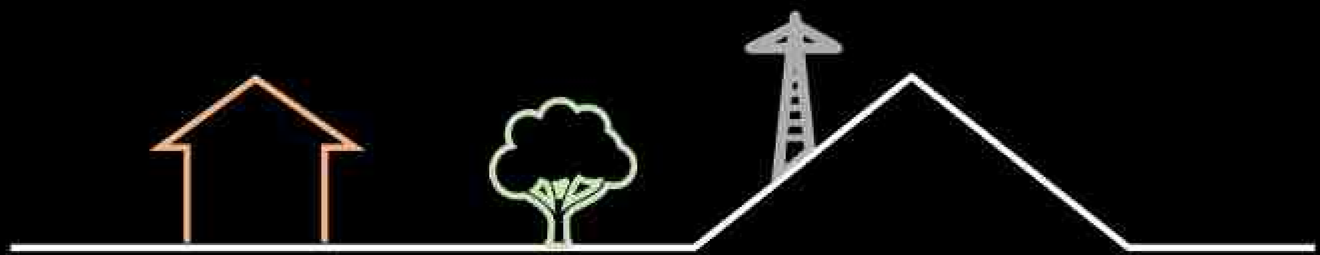
Spatial Data for Geography



Spatial Data for Geography

Land cover change

Military unit detection
Land use change
Disaster coverage
Agriculture and forestry
Ships and oil spills



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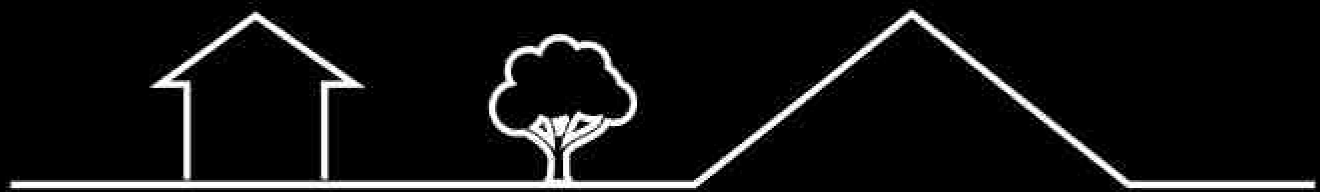

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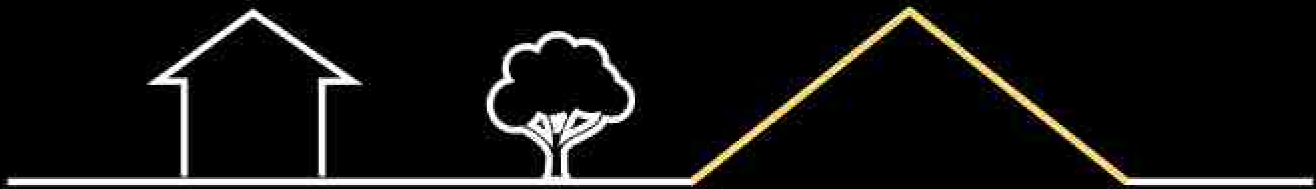

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Topographic change

Topographic analysis
Landslide volume
Mining volume
Building structure
Forester structure

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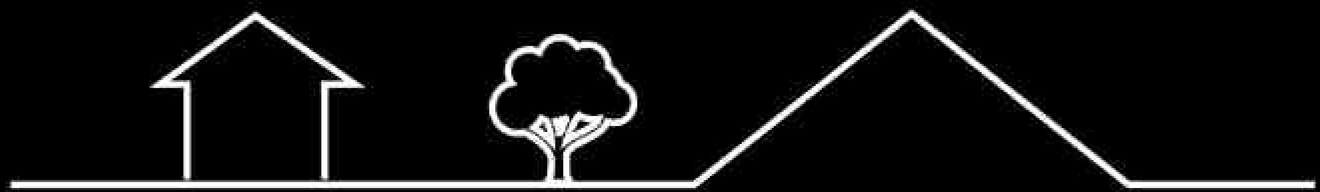

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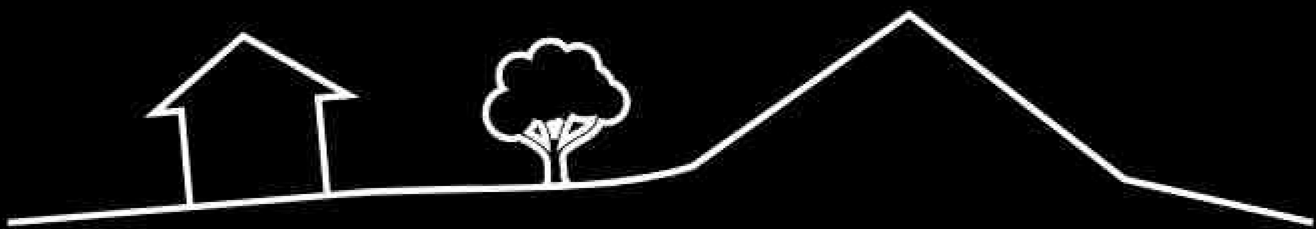

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Surface deformation

- Earthquakes
- Subsidence
- Surface creep
- Volcanoes
- Infrastructure

Why is geospatial data so special?

Specialty of geospatial Data

- Data types
- File format
- Coordinates
- Ontology
- Uncertainty
- Collection methods

Specialty of Geospatial Data

- Data types
- File format
- Coordinates
- Ontology
- Uncertainty
- Collection methods

- Vector
- Raster
- Attribute/tabular data
- Topography
- Spatiotemporal data

Specialty of Geospatial Data

- Data types
- File format
- Coordinates
- Ontology
- Uncertainty
- Collection methods

- Shapefiles
- Coverages
- Geodatabases
- Layer files
- Grids
- TINs/Mesh
- Point cloud
- Images/Geotiff
- Many more

Specialty of Geospatial Data

- Data types
- File format
- Coordinates
- Ontology
- Uncertainty
- Collection methods

- Projection
- Coordinate system
- Reference frame
- Transformation

Specialty of Geospatial Data

- Data types
- File format
- Coordinates
- Ontology
- Uncertainty
- Collection methods

臺大樹語
登入
首頁圖台
樹木搜尋
快速搜尋
M47
M52*小小福
M57*共同教學館
M58*農產品展售中心
舟山路
舟山路
舟山路
舟山路
椰林道
椰林道
S21*展書樓
S22*地質科學系
S23*鹿鳴堂
S24*禮賢樓
S41*生命科學館
S42*地理系館
S43*浩瀚樓
S44*氣象館
S46*精密溫室
地質系標本室
比例尺（1 : 846）
台灣TWD(304247.39214 , 2767570.98328)
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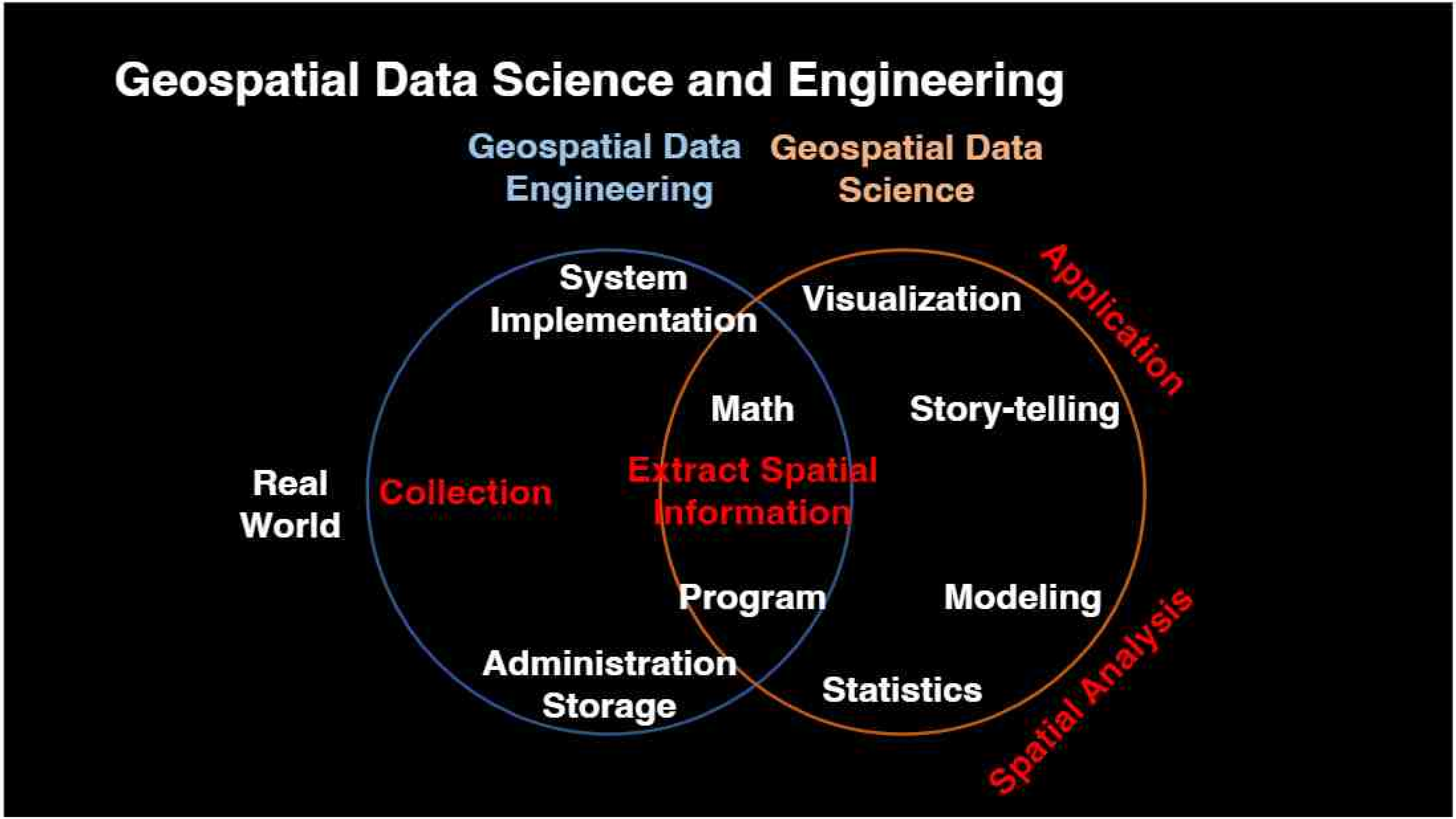
https://map.ntu.edu.tw/ntutree/index.htm


https://www.lidarsolutions.com.au/product/lidar-point-cloud-processing-software-lidar360-1-year-subscription/

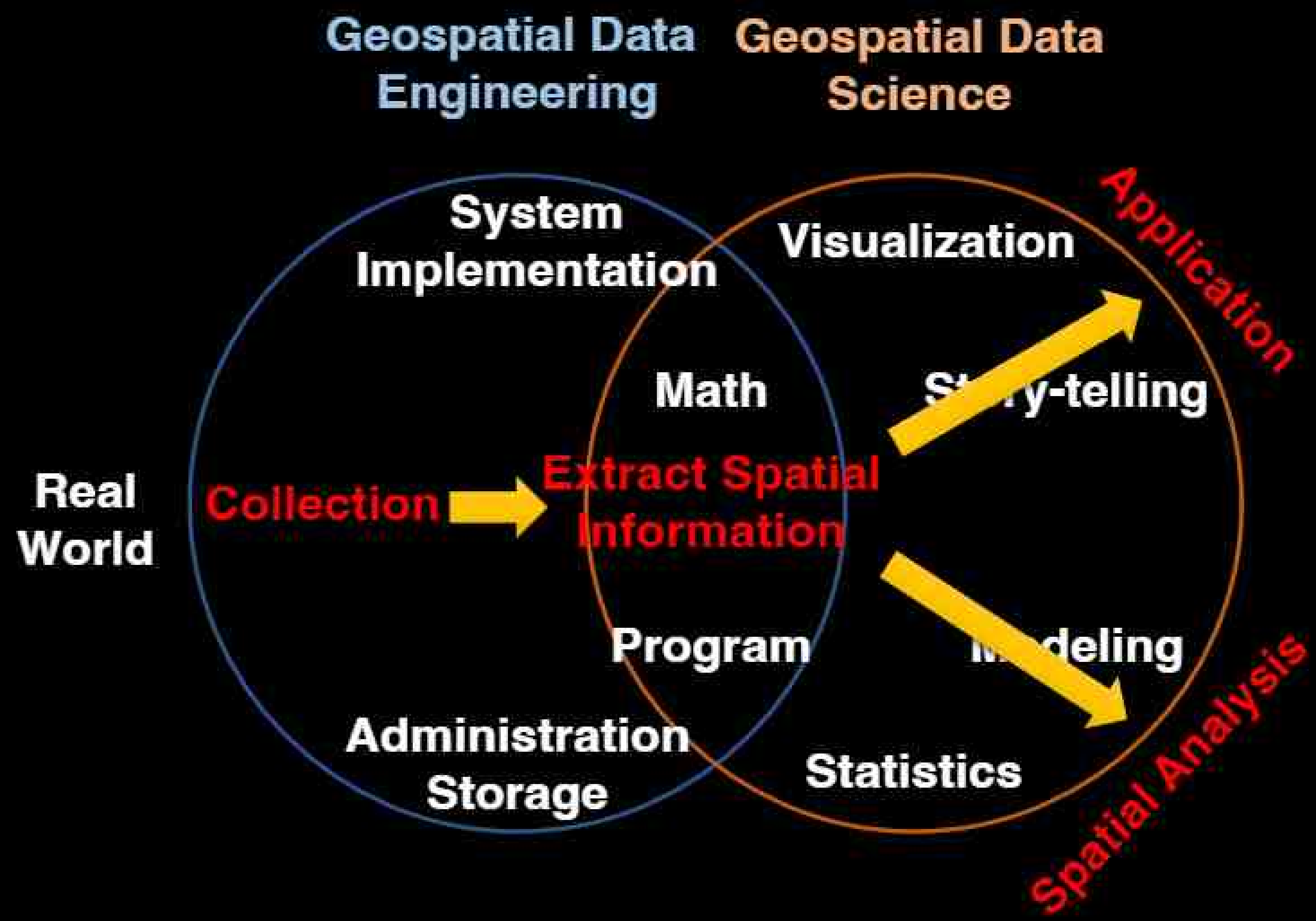
Specialty of Geospatial Data

- Data types
- File format
- Coordinates
- Ontology
- Uncertainty
- Collection methods

- Scale/resolution
- Positional accuracy
- Precision
- Age of data
- Metadata


Geospatial Data Science and Engineering
Geospatial Data Engineering
Geospatial Data Science
System Implementation
Visualization
Application
Math
Story-telling
Real World
Collection
Extract Spatial Information
Program
Modeling
Administration Storage
Statistics
Spatial Analysis

Geospatial Data Science and Engineering



What is geographic information?


https://www2.nhk.or.jp/archives/movies/?id=D0025010823_00000

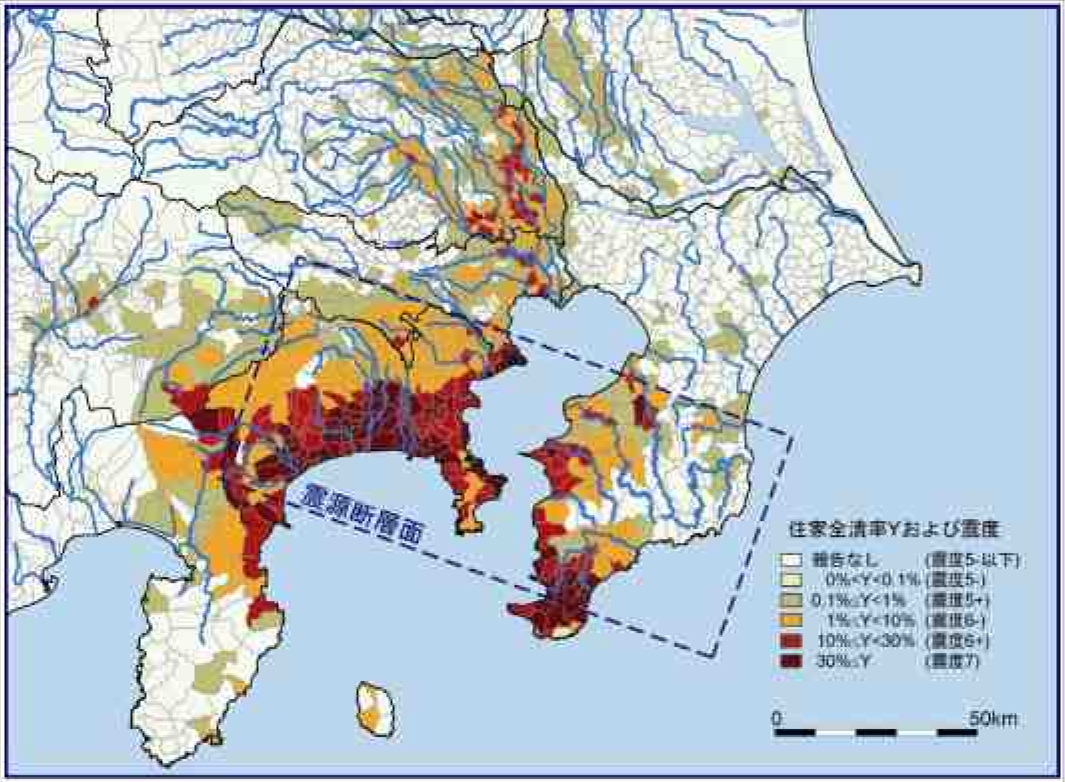
Questions You May Ask

- What is it?
  - 鎌倉大佛

Attribute data

- Where is it?
  - 日本神奈川縣鎌倉市
  - 高德院
  - 郵遞區號：248-0016
  - 電話區碼：467
  - 江之島電鐵長谷站出站後，沿著垂直鐵路的大路往右手邊走，步行約七分鐘
  - 35.31698080576993, 139.5357120099604

Spatial data



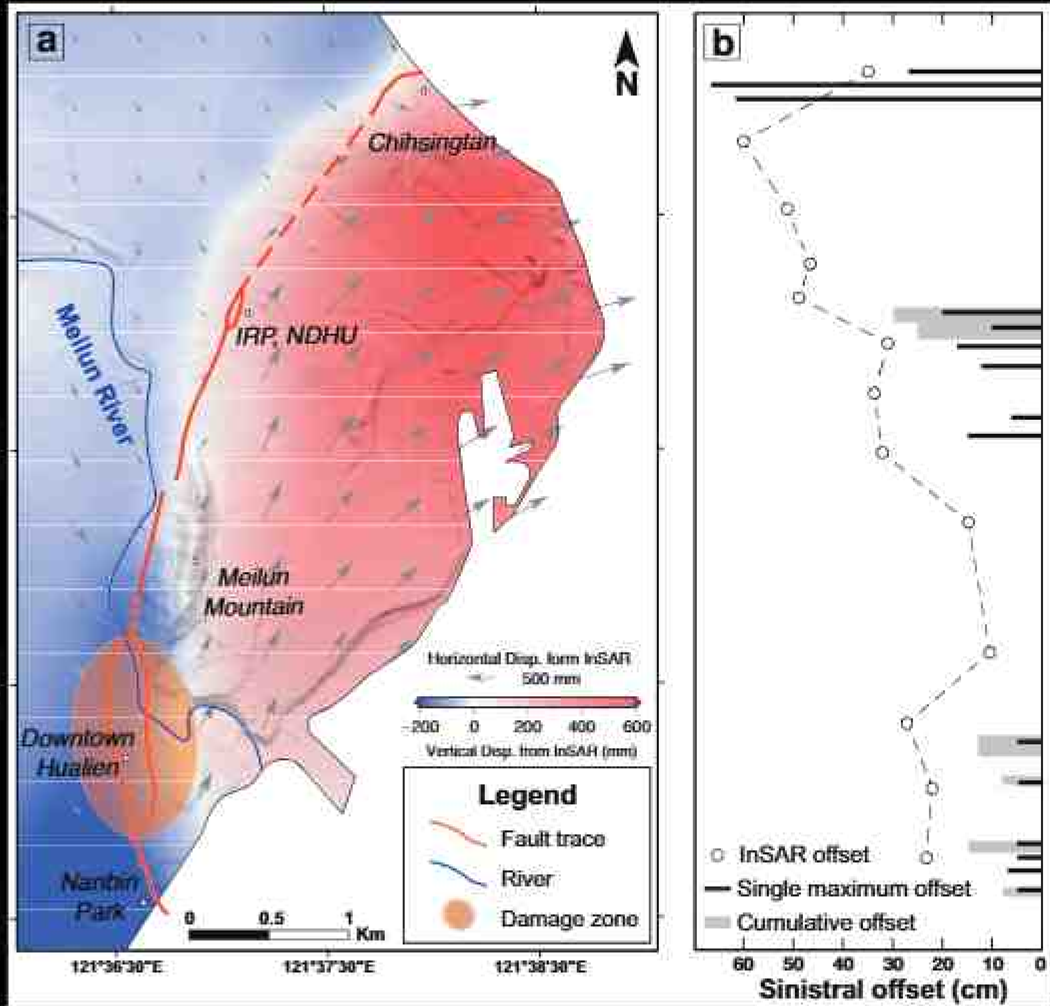
https://www.pref.kanagawa.jp/docs/j8g/100th/photo.html

諸井・武村 (2002)

Support Infrastructure for Decision-Making

Decision-making support infrastructure	Ease of sharing with everyone	GIS example
Wisdom ↑	*Impossible*	Policies developed and accepted by stakeholders
Knowledge ↑	*Difficult, especially tacit knowledge*	Personal knowledge about places and issues
Evidence ↑	*Often not easy*	Results of GIS analysis of many data sets or scenarios
Information ↑	*Easy*	Contents of a database assembled from raw facts
Data	*Easy*	Raw geographic facts

Longley et al. (2014)

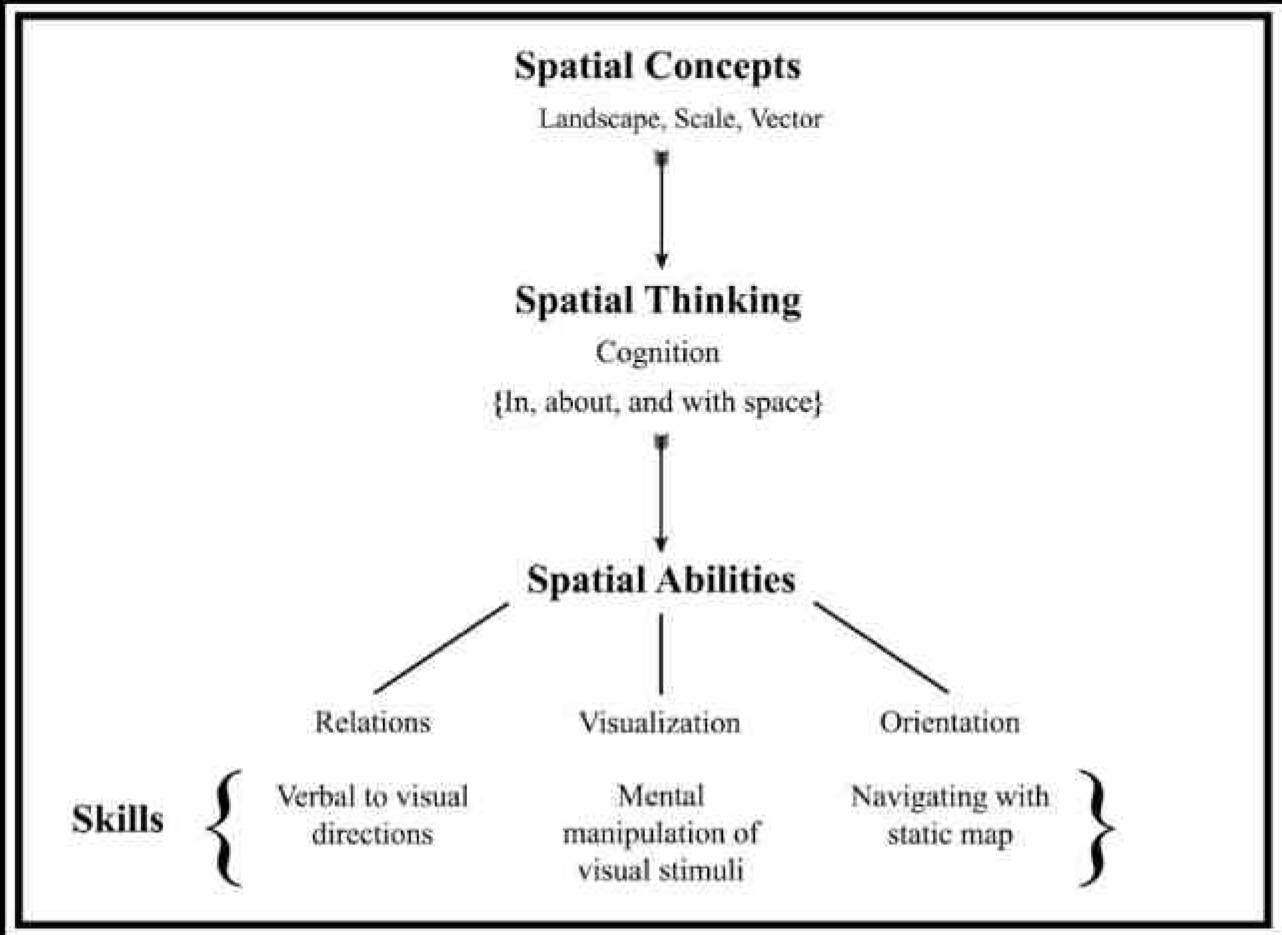


Lin et al. (2019)

Components of Geographic Information

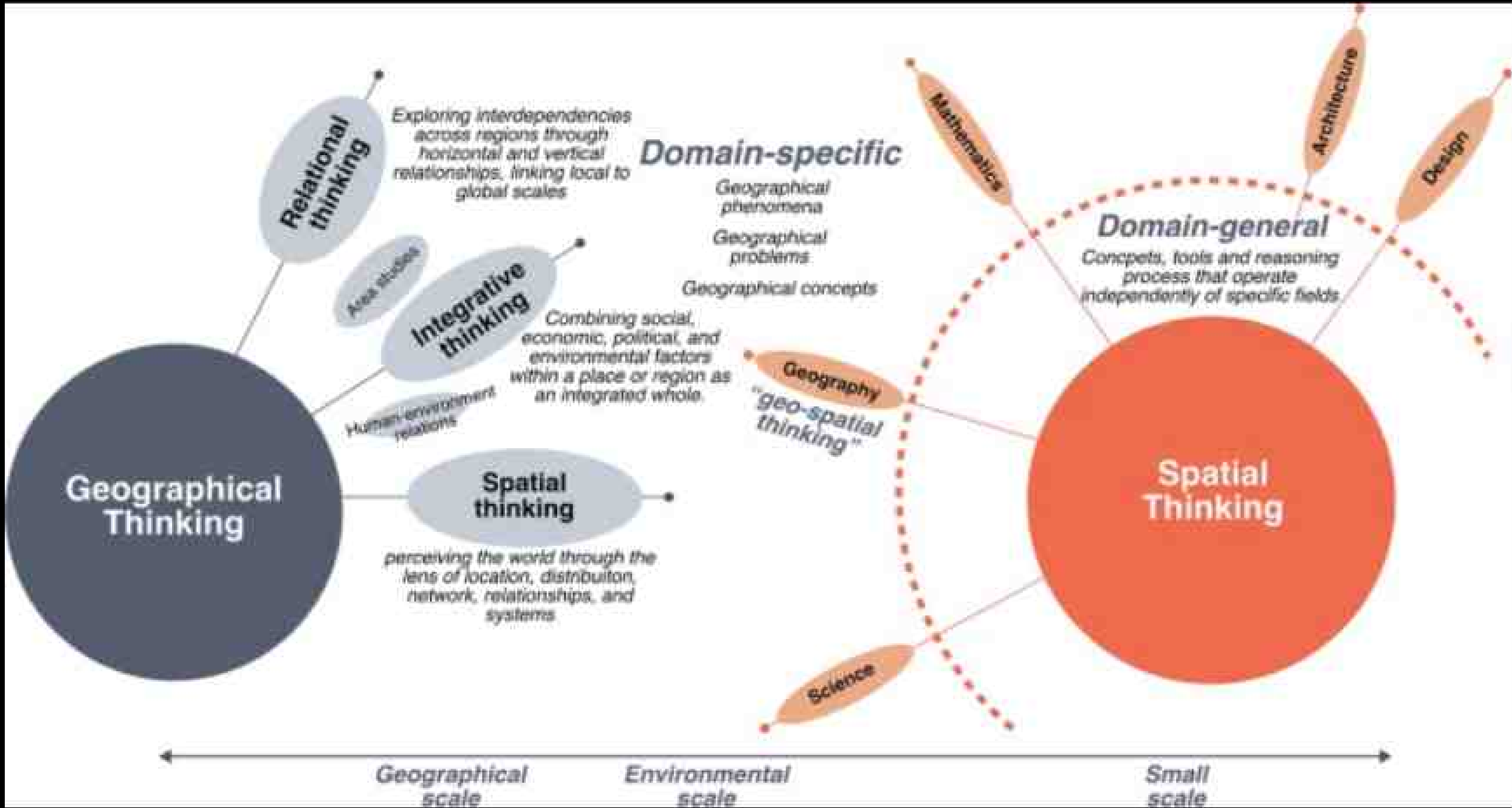
- Space/Location
  - Three-dimensional spot in real world
  - Spatial expanse is two-dimensional for most maps
- Attribute
  - Physical properties or aesthetic judgments
- Time
  - Most common map is a form of snapshot
    - Valid for specific point of time
  - Inexorable progression, inability to turn back

Spatial Thinking



Tomaszewski et al. (2020)

Geographic Thinking (Geo-thinking)



Lee (2024)

Homework 1 – Due 2PM, March 20

- A pdf file includes 1 Map + 1-page report.
- Upload a picture of a map, which you see in your daily life.
- Provide basic information of this map. Where is it? When did you take a picture of it or access it?
- Provide the title of the map, if any. Speculate the purpose of this map (from the map-maker's point of view).
- Describe geographic features on this map.
- Provide geographic thinking from the map.